

MacSync-WRTDX2 Wi-Fi RTD (PT100) Temperature Sensor Datasheet

Product Overview

MacSync-WRTDX2 is a high-accuracy **Wi-Fi-based RTD (PT100) temperature sensor** designed for industrial applications where precision, reliability, and direct IP connectivity are required. It delivers stable and accurate temperature measurements using **PT100 sensors with IEC 60751 linearization**, making it ideal for process monitoring and critical thermal applications.

Engineered for harsh industrial environments, MacSync-WRTDX2 features a rugged enclosure with industrial-grade electronics, surge protection, and optional IP-rated protection suitable for pipelines, machinery, furnaces, boilers, cold-chain systems, and manufacturing process lines.

The device transmits temperature data securely over standard **Wi-Fi networks (2.4 GHz / 5 GHz)** to **on-premise servers, SCADA systems, or cloud platforms** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**—without the need for gateways or long-range wireless infrastructure.

MacSync-WRTDX2 supports **event-based alerts, configurable sampling intervals**, and **local BLE-based provisioning** using the Macnman Maya mobile application. With flexible **battery-operated or DC-powered variants**, it is well suited for both fixed installations and locations where wired connectivity is limited.

From **industrial process control and predictive maintenance** to **HVAC monitoring, thermal safety compliance, and food & pharmaceutical temperature assurance**, **MacSync-WRTDX2 provides accurate, low-latency Wi-Fi temperature monitoring** for modern industrial IoT deployments.





Key Features

High-Accuracy PT100 RTD Measurement

Supports PT100 RTD probes with IEC 60751 linearization for precise and repeatable industrial temperature sensing.

Wi-Fi Based Connectivity

Direct temperature data transmission over standard Wi-Fi networks (2.4 GHz / 5 GHz) without gateways.

Industrial-Grade Rugged Design

IP-rated enclosure with resistance to dust, moisture, vibration, and harsh operating conditions.

Standard IP Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with SCADA systems, dashboards, and cloud platforms.

Local BLE Configuration

Easy provisioning, diagnostics, and parameter configuration using the Macman Maya mobile application.

Wide Temperature Measurement Range

Suitable for extreme low and high temperature processes across industrial and commercial environments.

Battery or DC Powered Operation

Available in battery-operated and externally powered variants for flexible industrial deployments.

Event-Based & Interval Temperature Alerts

Configurable threshold and time-based alerts for overheating, freezing, and process deviations.

Secure Wi-Fi Communication

WPA2 / WPA3 Wi-Fi security with TLS-encrypted data transmission for secure industrial communication.

Predictive Maintenance Ready

Enables early detection of abnormal temperature trends in motors, pumps, pipelines, and heat-critical assets.

Suited Applications

Industrial Process Temperature Monitoring

Continuous and high-accuracy PT100 temperature monitoring for chemical processing, metal fabrication, and manufacturing lines.

Boilers, Furnaces & Heaters

Reliable high-temperature monitoring for combustion control, thermal safety, and regulatory compliance.

Cold-Chain & Refrigeration Systems

Temperature assurance for pharmaceutical storage, food processing units, cold rooms, and refrigerated warehouses.

Food, Pharma & Clean-Room Compliance

Ensures temperature compliance in regulated environments such as pharma manufacturing, labs, and clean rooms.

Predictive Maintenance of Motors & Pumps

Early detection of overheating in motors, bearings, pumps, and compressors to prevent failures and unplanned downtime.

Pipelines & Industrial Assets

Accurate temperature measurement for pipelines, heat exchangers, and industrial assets within Wi-Fi-covered facilities.

HVAC & Building Automation

Integration with HVAC and BMS systems for thermal control, energy optimization, and indoor comfort monitoring.

Smart Factories & Industrial IoT

Enables real-time temperature visibility across machines and processes using direct Wi-Fi connectivity.

Technical Specifications

General Specifications

Part Order Number	MacSync-WRTDX2
Product Brand	Macnman Technologies Pvt Ltd

RTD Measurement

Sensor Type	RTD - PT100
Temperature Range	−200 °C to +850 °C
Accuracy	±0.2 °C (typical)
Resolution	0.1 °C
Repeatability	±0.05 °C
Sensor Linearization	IEC 60751 Standard Curve
Self-Heating Error	< 0.1 °C @ 1 mA excitation

Drift (1 year)	< 0.05 °C
Insulation Resistance	> 100 MΩ @ 500 VDC
Probe Options	Internal & External Probes Available
Probe Material	SS304 / SS316 (optional)

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers
	Private Cloud Deployments
	Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core
	Microsoft Azure IoT Hub
	Google Cloud (via MQTT / HTTPS)
	Custom IoT Platforms
Integration Methods	MQTT Brokers
	REST / HTTPS APIs
	Webhooks
Enterprise Systems	SCADA Systems
	BMS / IBMS Platforms
	Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode
	~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS

Serial Interface	RS485 Modbus RTU support (optional)
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Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WRTDX2 Variants

VARIANT NAME	RTD TYPE	POWER OPTION	IDEAL USE CASE
MacSync-WRTDX2-PO	PT100	12–24 V DC Power	High-precision temperature monitoring in automation systems
MacSync-WRTDX2-BO	PT100	Battery (Multi-year)	Cold-chain, pipelines & predictive maintenance monitoring

What's in the Box?

Make sure box contains following items before installations



MacSync



Mounting Clip



User manual & Warranty Card



Junction Box*



Mounting Screws

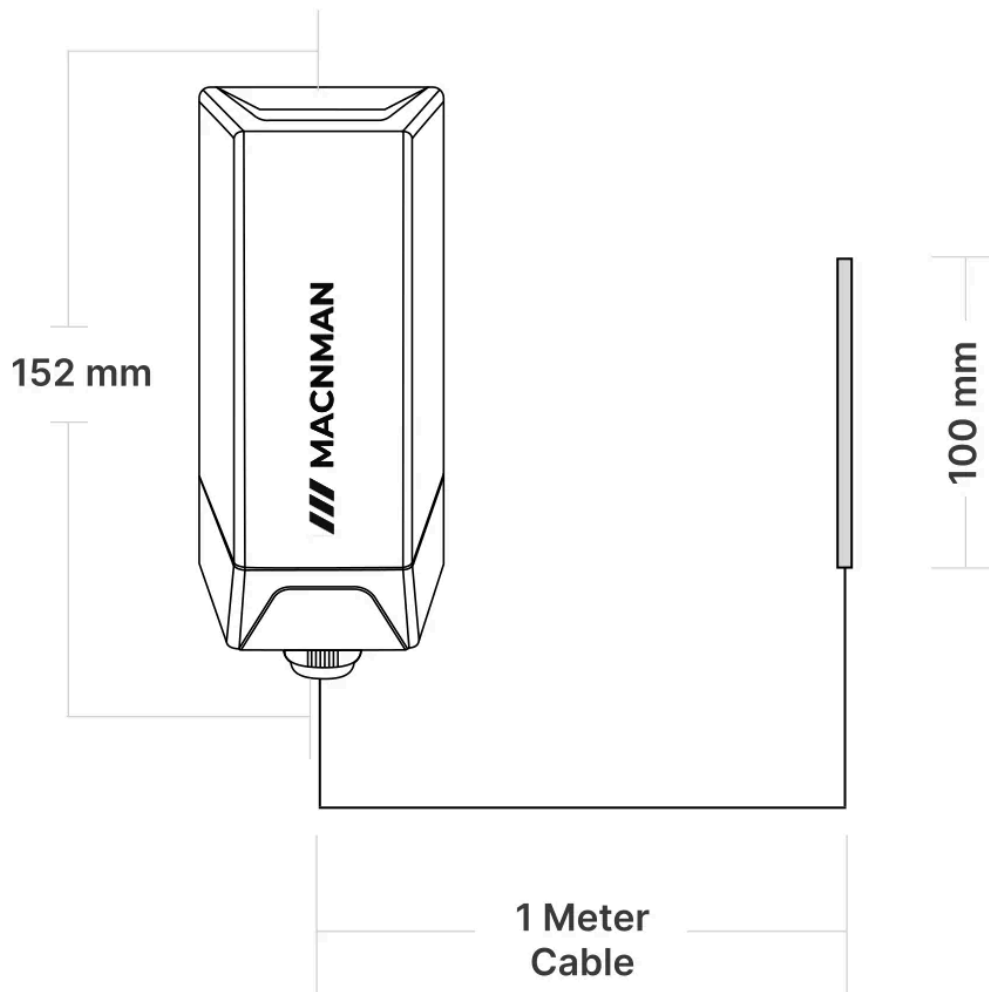


Power Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacSync-WRTDX2 Dimensions



Frequently Asked Questions

What is the MacSync-WRTDX2 used for?

MacSync-WRTDX2 is a high-accuracy Wi-Fi RTD (PT100) temperature sensor for industrial applications that need precise, reliable temperature monitoring. Typical applications include Industrial Process Temperature Monitoring, Predictive Maintenance of Motors & Pumps, Boilers, Furnaces & Heaters, Pipelines & Industrial Assets, Cold-Chain & Refrigeration Systems and HVAC & Building Automation.

What does the MacSync-WRTDX2 measure, and how accurately?

Measurement specifications of the MacSync-WRTDX2:

- Temperature: range $-200\text{ }^{\circ}\text{C}$ to $+850\text{ }^{\circ}\text{C}$.
- RTD: accuracy $\pm 0.2\text{ }^{\circ}\text{C}$ (typical), resolution $0.1\text{ }^{\circ}\text{C}$.

How does the MacSync-WRTDX2 connect and send data over Wi-Fi?

The MacSync-WRTDX2 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band). Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-WRTDX2 powered?

The MacSync-WRTDX2 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-WRTDX2?

The MacSync-WRTDX2 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to $+80^{\circ}\text{C}$. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WRTDX2 configured?

The MacSync-WRTDX2 is configured with the Macman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WRTDX2 work with?

The MacSync-WRTDX2 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacSync LRTD X1 LoRaWAN RTD temperature sensor](#), [MacSync-WRTDX5 Wi-Fi RTD PT1000 temperature sensor](#), [MWTH 301 compact Wi-Fi temperature and humidity sensor](#), [MacSync-WT1 Wi-Fi temperature and humidity sensor](#)
- Product page: [MacSync-WRTDX2 on macnman.com](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)